CSE430 DSP Lab03

Assignment 02

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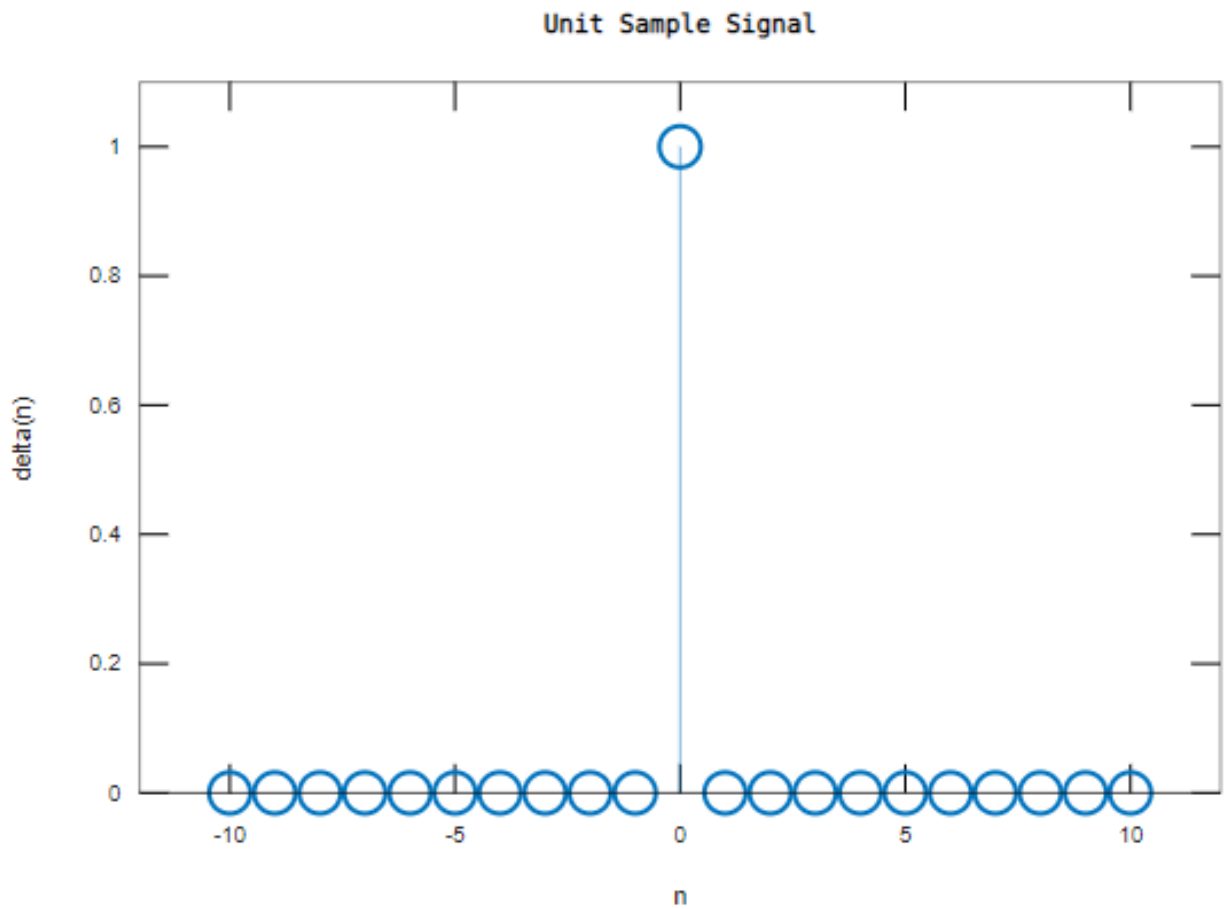
Task01 Unit Sample Signal

unit_sample_signal.m

RUN

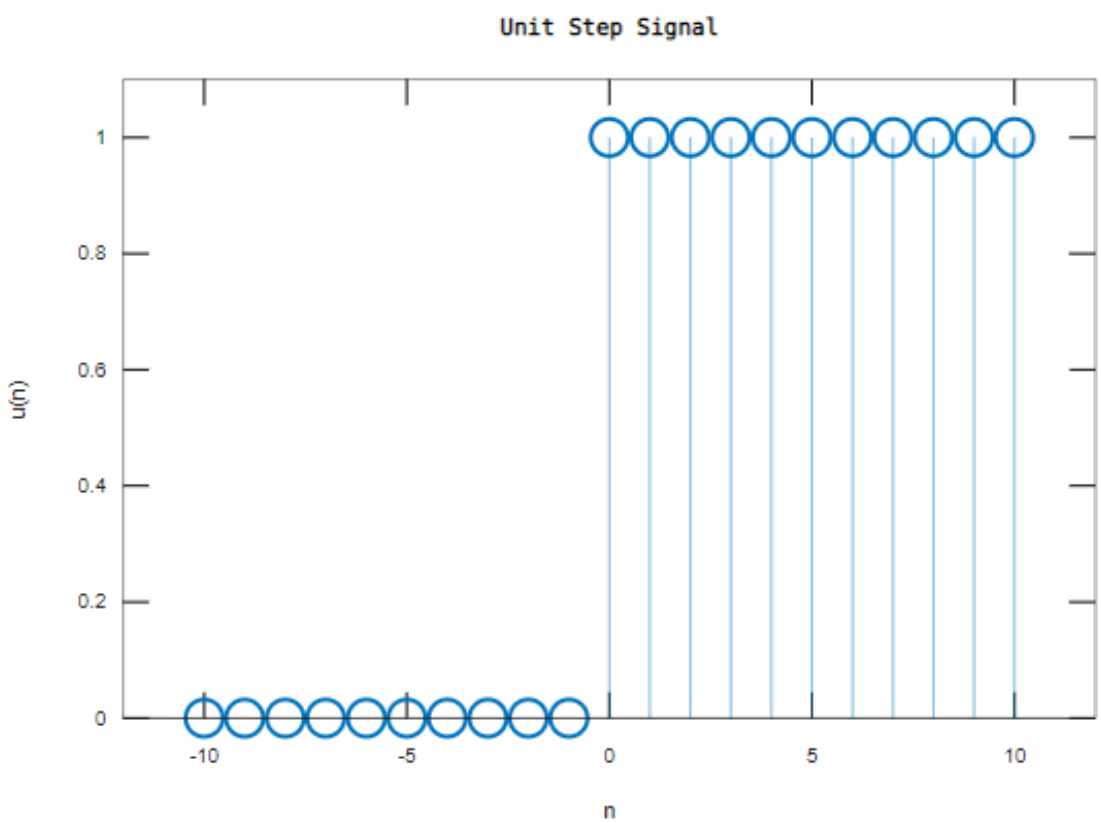
```
clc; clear all; close all;

n = -10: 10;
f = (n==0);
stem(n,f)

xlabel('n');
ylabel('delta(n)');
title('Unit Sample Signal');
axis([-12 12 0 1.1])
```


Unit Sample Signal
delta(n)
1
0.8
0.6
0.4
0.2
0
-10
-5
0
5
10
n

Task02 Unit Step Signal

unit_step_signal.m RUN ▶

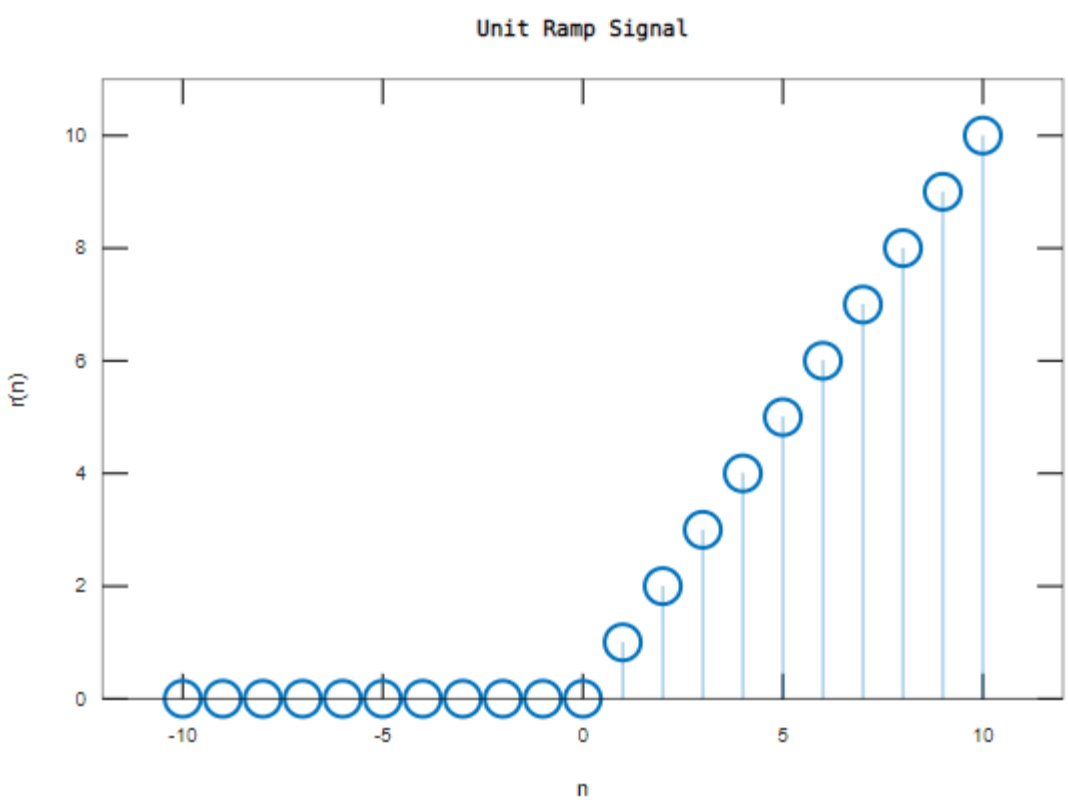
```
clc; clear all; close all;

n = -10: 10;

f=(n >= 0);
stem(n,f)

xlabel('n');
ylabel('u(n)');
title('Unit Step Signal');
axis([-12 12 0 1.1])
```



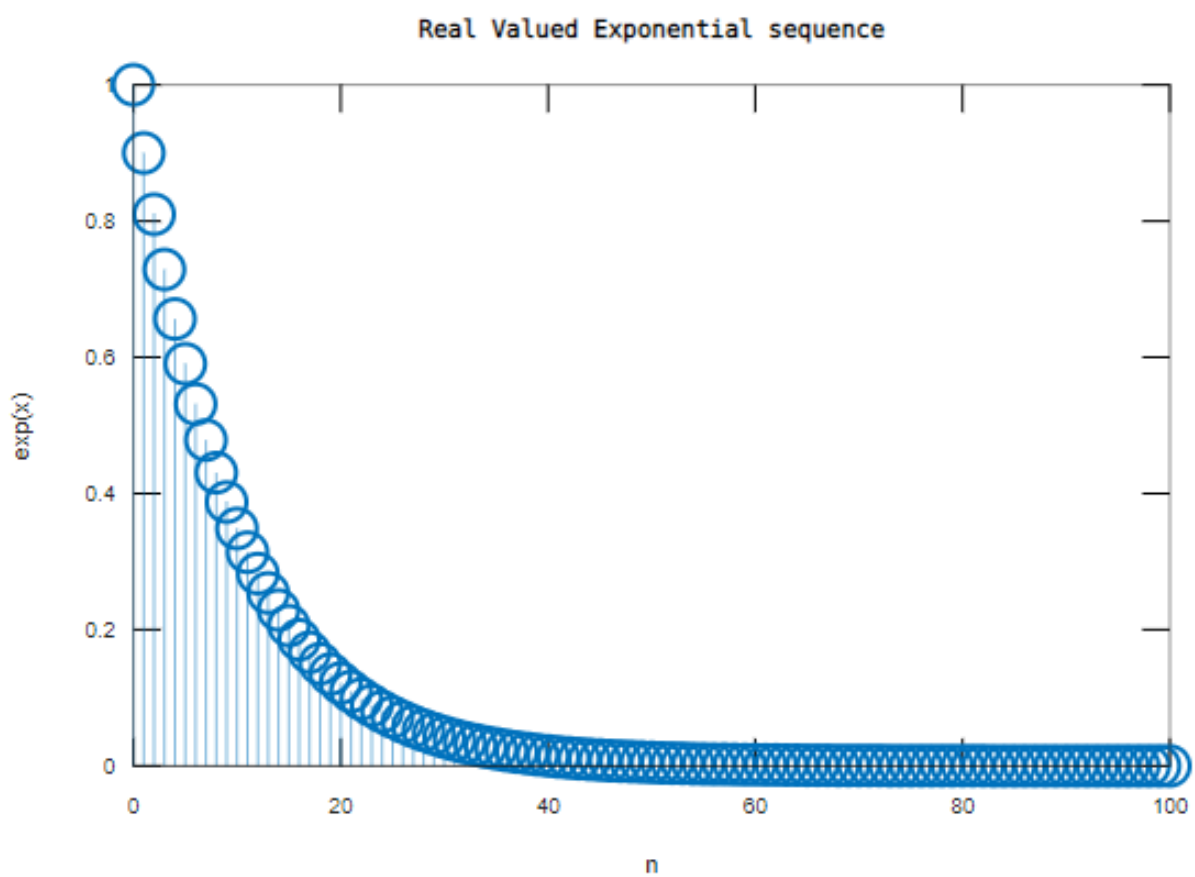
Task03 Unit Ramp Signal

unit_ramp_signal.m RUN ▶

```
clc; clear all; close all;

n= -10: 10;
[a,b] = size(n);
r = zeros(a,b);

for i = 1:b
    if n(i) >= 0
        r(i) = n(i);
    else
        r(i) = 0;
    end
end

stem(n,r)
xlabel('n');
ylabel('r(n)');
title('Unit Ramp Signal ');
axis([-12 12 0 11])
```



Task04 Real Valued Exponential sequence

real_valued_exponential.m RUN ▶

```
clc; clear all; close all;

n = 0:100;
x = (0.9).^n;

stem(n,x)

xlabel('n');
ylabel('exp(x)');
title('Real Valued Exponential sequence');
```